

Setup ($O(l_p^2 2^{l_p})$ time)
check every proof $b \in \mathbb{B}^{l_p}$;
keep valid $\dot{\pi}$ with $\ell(\dot{\pi}) \leq \tilde{l}$
 $\Rightarrow$ **set** Π_{VA}

Truncate: force $\dot{\pi}$ to output
 $(a, 0)$ if it has not halted
by time $\tilde{t}$ (ensures $t(\dot{\pi}) \leq \tilde{t}$)

For cycle $k = 1, 2, 3, \dots$: **run every** $\dot{\pi} \in \Pi_{VA}$ **on** $h_{<k}$,
obtaining $(a_k^{\dot{\pi}}, v_k^{\dot{\pi}}) = \dot{\pi}(h_{<k})$

Pick $\dot{\pi}^* = \arg \max_{\dot{\pi} \in \Pi_{VA}} v_k^{\dot{\pi}}$; **act** $a_k = a_k^{\dot{\pi}^*}$; **observe** e_k

Per-cycle cost $O(t_{\text{cycle}}) = O(2^{\tilde{l}} \cdot \tilde{t})$ — independent of the number of elapsed cycles k

loop